Acquiring hidden algebraic response types

Exact counting and optimal synthesis from manufactured summaries

Astra P3 research

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Abstract

We construct an exact representation from a raw dense Boolean polynomial. Its higher derivatives reveal primitive coordinate blocks; a metric on quadratic block responses then acquires unknown local changes of basis and repeated response types. The output is a symmetric matrix of rank at most R, an explicitly retained residual with at most $15\beta^2(2E+n)$ scalar edges, and local nonlinear terms. Here β is the largest acquired block, R is the total dimension of the hidden response types, and E is the sum of the ranks of the hidden off-block errors. Neither the blocks, bases, types, errors nor type profile are supplied. Acquisition costs $2^{O(\beta^2)}\,\mathrm{poly}(L+n)$ time and polynomial space; for fixed block sizes it is polynomial.

Combined with the earlier local-phase extension of the pinned OpenAI counting manuscript, this gives exact counting in $\mathrm{poly}(L+n)2^{o(n)}$ time when $\beta = n^{o(1)}$, $R = o(n)$, and $\log(2+E/n) = o(\log n)$. It removes the earlier rank-times-error condition for this structured family, while preserving the original independent input count. The acquired coordinate classes also yield shared $\mathbb{Z}/4\mathbb{Z}$ summaries. Their exact labelled fiber populations permit optimal Boolean prediction within this acquired feature class, with a certified uniform-input error count, at the same subexponential scale.

Acquisition, identities and restricted failure theorems are unconditional. The subexponential counting and synthesis bounds depend on the governing manuscript. Classical centroid, clustering, co-clustering, sum-rank and Gaussian ingredients are credited. No historical first-priority, generic-circuit, polynomial-time SAT, or practical speed claim is made. Executed programs validate acquisition and exact reconstruction with exhaustive finite leaf counters.

1 Input and precise guarantees

All matrix ranks are over $\mathbb{F}_2$. For an alternating matrix B (symmetric with zero diagonal), put

$$q_B(x) = \sum_{i<j} B_{ij}x_ix_j, \qquad e(B) = |\{(i,j) : i < j,\ B_{ij} = 1\}|.$$

The initial input is a degree-at-most-three Boolean polynomial p in explicit algebraic normal form (ANF), on n independent bits. The bit description length L includes every coefficient and index; it may contain $\Theta(n^3)$ terms. The targets are

$$Z(p) = \sum_{x\in\mathbb{F}_2^n} (-1)^{p(x)}, \qquad N_\tau(p) = \frac{2^n + (-1)^\tau Z(p)}{2}.$$

Let $T(u,v,w) = D_uD_vD_wp(0)$ be its third additive derivative. We first require that T have zero radical. Its primitive direct-sum blocks are acquired in polynomial time below. Write their dimensions as $b_1,\dots,b_N$, with $\sum_a b_a = n$ and $\beta = \max_a b_a$. The useful promise concerns the quadratic interactions in *some unknown bases* of these blocks:

$$B_{aj} = L_aK_{\tau(a),\tau(j)}L_j^\top + S_{aj} \quad (a \neq j). \tag{1}$$

Here $L_a \in \mathrm{GL}_{b_a}(\mathbb{F}_2)$, a type can be shared only by blocks of the same dimension, and K is an arbitrary symmetric matrix indexed by type-coordinate pairs. If c_d types have dimension d, put

$$R_* = \sum_d dc_d, \qquad E = \sum_{a<j} \operatorname{rank} S_{aj}.$$

The within-block quadratic terms are arbitrary. The compiler is given none of τ, L_a, K, S, or the profile (c_d). Independent block basis changes preserve E. If the hidden error has M scalar edges, $E \le M$; if it occupies M_B block pairs, $E \le \beta M_B$. Arbitrary affine disguises of the original inputs are allowed: their effects on lower-degree terms are charged below.

Theorem 1.1 (Matrix acquisition)**.** *Given a raw alternating matrix B, a partition into blocks of sizes b_a, and an integer budget $0 \le R \le n$, a deterministic algorithm using $2^{O(\beta^2)}\,\mathrm{poly}(n)$ time and polynomial space returns a verified block-diagonal invertible G, a partition of scalar coordinates into $d \le R$ nonempty classes, a symmetric d by d matrix H, and an alternating residual S' such that*

$$GBG^\top = \mathrm{off}(C) + S', \qquad C = UHU^\top, \qquad \operatorname{rank} C \le d \le R.$$

The columns of U are the class indicators; each row has one 1. A budget too small for one type of each occurring block dimension is reported infeasible. If (1) holds with $R_* \le R$,

$$e(S') \le 15\beta^2(2E + n). \tag{2}$$

No coordinates are deleted. Recovering the planted witness is unnecessary.

Theorem 1.2 (Raw polynomial counting)**.** *Assume a raw cubic p has radical-free third derivative and satisfies (1) with $R_* \le R$. Acquire its primitive blocks, then apply Theorem 1.1. The algorithm constructs an invertible map $x = Ay$ and the exact identity*

$$p(Ay) = q_{\mathrm{off}(C)}(y) + q_{S'}(y) + f(y), \tag{3}$$

where every monomial of f is inside an acquired block and has degree at most three. Its nonconstant monomial count is $m_0 = O(n\beta^2)$. Conditional on the local-phase theorem in Section 5, the exact counting bit time is

$$\mathrm{poly}(L+n)\left(2^{O(\beta^2)} + 2^{R+F(n,m_0+15\beta^2(2E+n)+O(n))}\right), \tag{4}$$

with polynomial space. It is $\mathrm{poly}(L+n)2^{o(n)}$ whenever

$$\beta = n^{o(1)}, \qquad R = o(n), \qquad \log(2 + E/n) = o(\log n). \tag{5}$$

The exponent uses the original n independent inputs.

This does not cover arbitrary dense cubics. The favorable intrinsic structure is restricted, but its representation is acquired and checked. Failed tests or unfavorable parameters permit exhaustive fallback. All budgets $R \le n$ can be inspected with polynomial overhead. Choose a certificate by an explicit guarded counting bound or dovetail its exact counter with the others. Sequentially exhausting a bad exponential case before trying a good one would not justify this bound.

2 Acquiring the nonlinear blocks

Centroid decomposition is established multilinear-form theory [2]; its use against hidden block-diagonal cubic descriptions also appears in [3, Appendix A]. We give the binary construction to make characteristic and costs explicit. Compute

$$\mathcal{A} = \{P : T(Pu, v, w) = T(u, Pv, w) = T(u, v, Pw) \text{ for all } u, v, w\}.$$

Basis-vector tests give $2n^3$ linear equations in n^2 binary unknowns. For $P, Q \in \mathcal{A}$, slot transfers show

$$T(PQu, v, w) = T(Qu, Pv, w) = T(u, Pv, Qw) = T(Pu, v, Qw) = T(QPu, v, w).$$

The zero radical implies $PQ = QP$. Transfers also give closure under multiplication and the identity lies in $\mathcal{A}$. Thus $P \mapsto P^2 + P$ is linear on this commutative binary algebra. Its kernel contains exactly all idempotents, obtained by another linear system without assuming semisimplicity.

Starting with I, split each current projector P into its nonzero pieces PE and $P(I + E)$ for every basis idempotent E. The pieces commute and are orthogonal; at most n survive. A further idempotent is a linear combination of the processed basis and acts on every final piece as zero or identity, so no further split exists. Moreover

$$T(Ex, (I + E)y, z) = T(x, E(I + E)y, z) = 0.$$

Conversely, a direct-sum projection satisfies the centroid equations. The resulting images are therefore the intrinsic primitive blocks, up to permutation. Their chosen bases give an invertible column matrix V. Naive elimination costs, for example, $O(n^7)$ binary operations and polynomial space.

The compiler substitutes into the *full ANF* $p(Vu)$. Transforming only the quadratic coefficient matrix is incorrect: Boolean substitution into a cubic can create quadratic and unary terms. In the acquired coordinates these corrections stay within blocks. The change between any hidden and acquired block bases is block-diagonal up to permutation, so (1) and E remain valid, with arbitrary new diagonal blocks. Affine translations also create only local lower-degree terms from the local cubic parts, and unary terms from the quadratic part.

If T has a radical, the earlier Gaussian-radical compiler can return an exact smaller core. The present theorem then applies only if that core has the stated properties. Eliminating a large radical is not assumed to preserve the response types.

3 An invariant metric and acquisition of the type profile

For two d by n row blocks, divided into d by b_j pieces, put

$$\rho_d(X, Y) = \sum_j \operatorname{rank}(X_j - Y_j), \qquad \bar{\rho}_d([X], [Y]) = \min_{Q \in \mathrm{GL}_d(\mathbb{F}_2)} \rho_d(QX, Y).$$

Rank subadditivity makes ρ_d a metric. Left invertible multiplication is an isometry, so $\bar{\rho}_d$ is a metric on orbits, or a pseudometric on raw row blocks. Symmetry uses Q^{-1} and the triangle inequality composes two gauges. Independent right invertible changes on the column blocks are also isometries. These are elementary sum-rank properties [6]; scalar Hamming distance would not suffice before the unknown bases are aligned.

Complete the ideal matrix from (1) on its diagonal blocks using the same symmetric kernel, obtaining D. Each arbitrary diagonal-block discrepancy has rank at most b_a. Consequently the total directed row-block error satisfies

$$T_0 := \sum_{a,j} \operatorname{rank}(B_{aj} - D_{aj}) \le 2E + n. \tag{6}$$

After left normalization, ideal row blocks of a common type are identical. Let T_d be the error contribution of dimension d. In each hidden type choose the observed row closest to its ideal center. If the errors are ϵ_a, distance to this observed center is at most $\epsilon_a + \min \epsilon_a$. Summing proves that the optimum with c_d *observed* centers costs at most $2T_d$.

For every occurring dimension d and every $c = 1, \ldots, N_d$, run exact one-swap metric c-median local search. Its established factor-five guarantee [4] gives

$$\kappa_d(c_d) \le 10T_d$$

for the promised budget. Distances are integral and at most n, so objectives are at most $N_d n$; each strict improvement lowers this by at least one. Thus exact local search terminates after polynomially many swaps. Zero-distance duplicates and discarded empty facilities do not invalidate the guarantee.

The hidden profile is selected by a polynomial budget DP:

$$\min \sum_d \kappa_d(c_d) \quad \text{subject to} \quad \sum_d d c_d \le R, \quad 1 \le c_d \le N_d.$$

Only occurring dimensions enter. The promised profile is feasible, so the chosen solution obeys

$$\kappa \le 10 \sum_d T_d \le 10(2E + n). \tag{7}$$

The DP needs neither E nor monotonicity of the acquired costs. It has polynomially many integer states, transitions and backpointers.

All orbit distances and minimizing gauges are computed by streaming fewer than 2^{d^2} binary matrices and testing invertibility. Only the best gauge for each pair is stored; their total bits are at most $\sum_d d^2 N_d^2 \le n^2$, apart from indexing. For uniform blocks a coarse distance cost is $O(2^{d^2} n^3)$; unequal blocks and all budget searches still cost $2^{O(\beta^2)}\,\mathrm{poly}(n)$ time and polynomial space. Materializing the whole group for growing β would violate the space claim and is not required.

4 Simultaneous alignment and exact symmetric rounding

Let $s(a)$ be row block a's selected observed center and choose $G_a B_a$ closest to $B_{s(a)}$. Set $G = \mathrm{diag}(G_a)$, $B' = GBG^\top$, and define the generally nonsymmetric model

$$X_{aj} = B_{s(a),j} G_j^\top .$$

It repeats the *raw* center row followed by the common right transform. Using the center row of B' instead would incorrectly insert that center's own left gauge. Right invariance gives

$$\sum_{a,j} \mathrm{rank}(B'_{aj} - X_{aj}) = \kappa, \qquad A_0 := \mathrm{Ham}(B', X) \le \beta^2 \kappa.$$

The last bound holds because each nonzero rectangular block has at most β^2 nonzero entries and rank at least one. Scalar coordinates of the same acquired block type and the same position form one class. There are $d \le R$ such classes, and the rows of X are constant within them.

Lemma 4.1 (Symmetric rounding)**.** *If B' is symmetric and X is constant on each row class of a partition also used for columns, majority-rounding B' on each class-pair block, with consistent zero ties, gives a symmetric $C = UHU^\top$ with*

$$\mathrm{Ham}(B', C) \le 3\,\mathrm{Ham}(B', X).$$

Proof. Let $P(X)$ replace each row's entries on every column class by their majority. It is a best Hamming projection onto column-class-constant matrices and remains row-class constant. Since $X^\top$ is column-class constant,

$$\mathrm{Ham}(X, P(X)) \le \mathrm{Ham}(X, X^\top) \le \mathrm{Ham}(X, B') + \mathrm{Ham}(B', X^\top) = 2\,\mathrm{Ham}(B', X).$$

Thus $\mathrm{Ham}(B', P(X)) \le 3\,\mathrm{Ham}(B', X)$. Majority projection of B' does at least as well as this possibly nonsymmetric candidate. Symmetry and consistent ties give a symmetric majority table H; the class indicator matrix U yields $C = UHU^\top$. □

This principle has close co-clustering ancestry [5, Theorem 3.6]. Its role here is an exact interface with the acquired finite-field gauges. Retain C's free diagonal: zeroing it can destroy low rank. The residual $S' = B' + \mathrm{off}(C)$ is alternating, and off-diagonal errors count twice in Hamming distance. Hence

$$e(S') \le \tfrac{3}{2} A_0 \le 15\beta^2(2E + n).$$

Direct checks of symmetry, rank and every coefficient certify the identity regardless of whether the hidden witness exists. This proves Theorem 1.1.

The final coordinates satisfy $u = G^\top y$, $x = VG^\top y$. Recompute $p(VG^\top y)$ and subtract $q_{B'}$. Every cubic-induced quadratic correction stays within a block, and substitution into a quadratic creates only an additional unary term beyond its congruent polar matrix. Thus the difference is block-local and has $O(n\beta^2)$ monomials, proving (3).

5 Counting and full reconstruction costs

The earlier Astra P3 local-phase theorem, conditional on Proposition 3.4, Lemma 4.7 and Appendix B of the pinned source [1], counts m explicit fixed-locality phases of fixed order 2^b in $\mathrm{poly}(L + n)2^{F(n,m)}$ bit time and polynomial space. It has $F(n, m) = o(n)$ whenever $\log(2+m/n) = o(\log n)$. In its guarded small-parameter regime,

$$h = \lceil \log_2(2 + km/n) \rceil + O_{k,b}(\log\log n), \quad a = h/\log_2 n, \quad F = O_{k,b}\Big(n\sqrt{a}[\log_2(2/a)]^{3/2}\Big).$$

Outside the guards it uses exhaustive enumeration. Its filtered phase products use the algebra

$$\mathbb{Q}(\zeta)[V]/(V^{2^{b-1}(n+2)}).$$

They have retained input degree $O_{k,b}(n)$ independently of m. The source manufactures replacement forms and summaries. Exact reconstruction precedes the final coefficient-sum map modulo 2^{n+2}; that map is not a ring homomorphism obtained by substituting $V = 1$ into the truncated ring. The full earlier phase proof and audits accompany the source archive.

For the acquired $C = UHU^\top$, define

$$Q_H(z) = \sum_a H_{aa} z_a + 2 \sum_{a<b} H_{ab} z_a z_b \pmod 4.$$

Because each row of U is a unit vector, its exact refinement is

$$(-1)^{q_{\mathrm{off}(C)}(y)} = i^{Q_H(U^\top y)} \prod_j i^{-C_{jj} y_j}.$$

For general overlapping column codes an extra unary sign is required; it vanishes for these indicator classes. Walsh expansion yields

$$i^{Q_H(z)} = 2^{-d} \sum_{\alpha \in \mathbb{F}_2^d} \Gamma_H(\alpha)(-1)^{\alpha^\top z}, \qquad \Gamma_H(\alpha) = \sum_z i^{Q_H(z)}(-1)^{\alpha^\top z}.$$

Exact quadratic elimination computes each Gaussian integer $\Gamma_H(\alpha)$ in polynomial time and $O(d)$ bits. Thus each of at most 2^d characters changes only unary phases. The inner counter sees $m_0 + e(S') + O(n)$ factors of locality at most three and phase order four.

Stream the characters. Reconstruct each inner answer exactly in $\mathbb{Z}[i]$ before multiplying by Γ_H and dividing the final numerator by 2^d. Numerators use $O(n + d)$ bits; the final count is integral in $[-2^n, 2^n]$. There is no loss of precision before division. This proves Theorem 1.2.

5.1 Raw programs and original outputs

For an XOR/AND straight-line program, propagate exact Boolean ANFs, reducing $x_i^2 = x_i$ after every instruction. An intermediate degree-at-most-three function has $O(n^3)$ terms; multiplying two such functions uses at most $O(n^6)$ term pairs. Reject this front end or use a separate fallback if the reduced result exceeds degree three. Otherwise the intermediate-degree property is certified, with conservative cost $O(Sn^6)$ coefficient operations for S instructions and polynomial storage. Long affine computations and shared cubic computations therefore introduce no internal-state counting variables. This does not recognize every low-degree final output through arbitrary high-degree intermediate states.

Restricting acquired coordinates preserves the local foreground and the rank bound. Padding to the original n variables maintains the uniform counting estimate with an exact known multiplicity. Conditional counts consequently produce witnesses and uniform satisfying samples, mapped back by $x = Ay$. Self-reduction itself is established; the new scope is the acquired family.

6 Universal fibers and optimal Boolean synthesis

For the acquired scalar coordinate classes $A_1, \dots, A_d$, set

$$s_a(y) = \sum_{j \in A_a} y_j \pmod 4.$$

These summaries are linear over $\mathbb{Z}/4\mathbb{Z}$ on the acquired Boolean coordinates. In the original inputs, first compute the retained invertible binary coordinate change; they are not claimed to be $\mathbb{Z}/4\mathbb{Z}$-linear directly in the original bits. Since $\binom{w}{2} \bmod 2$ depends only on $w \bmod 4$,

$$q_{\mathrm{off}(C)}(y) = \sum_a H_{aa} \binom{s_a(y)}{2} + \sum_{a<b} H_{ab}(s_a(y) \bmod 2)(s_b(y) \bmod 2) \pmod 2. \tag{8}$$

Denote this sign by $h_H(s)$ and put $F_0(y) = (-1)^{f(y)+q_{S'}(y)}$, including its constant. The signed local fiber population satisfies

$$W(s) := \sum_{y : s(y)=s} F_0(y) = 4^{-d} \sum_{\chi \in (\mathbb{Z}/4\mathbb{Z})^d} i^{-\chi \cdot s} T_\chi, \quad T_\chi = \sum_y F_0(y) i^{\sum_a \chi_a s_a(y)}. \tag{9}$$

Character orthogonality proves this identity. Every T_χ has the same local factors and only additional unary fourth-root phases. It is an exactly reconstructible family shared by all fibers and by any subsequent Boolean function of the summaries.

Theorem 6.1 (Exact synthesis within acquired summaries)**.** *Under the hypotheses of Theorem 1.2 and (5), conditional on the local-phase theorem, the exact numbers of zeros and ones of the original p can be streamed for every summary fiber. One can output a Boolean predictor minimizing uniform-input mistakes among* all Boolean functions of these acquired summaries. *The bit time is*

$$\mathrm{poly}(L+n)\left(2^{O(\beta^2)} + 2^{4R+F(n,m_0+e(S')+O(n))}\right),$$

with polynomial working space and at most 4^R output rows.

Proof. Compute $a_j(t) = \sum_{w \equiv t \ (4)} \binom{|A_j|}{w}$ by four-state dynamic programming. The unlabelled fiber size is $N(s) = \prod_j a_j(s_j)$, because the classes partition all coordinates. Its original-label imbalance is $h_H(s)W(s)$. Consequently

$$N_0(s) = \frac{N(s) + h_H(s)W(s)}{2}, \qquad N_1(s) = \frac{N(s) - h_H(s)W(s)}{2}.$$

Predict one exactly when $N_1(s) > N_0(s)$, choosing zero on ties and empty fibers. The independent fiber optima give

$$\mathrm{Err}_{\min} = \sum_s \min(N_0(s), N_1(s)) = \frac{2^n - \sum_s |W(s)|}{2}. \tag{10}$$

There are 4^d fibers and 4^d characters for each. Recompute the T_χ as necessary and output each fiber immediately. This gives the time and polynomial working space. Each inner answer is reconstructed exactly before division by 4^d; the numerator has $O(n + 2d)$ bits. Populations and error totals have $O(n)$ bits. □

Here d is the number of indicator classes, not merely $\operatorname{rank} C$. Compacting to an overlapping column basis loses the product population formula. The possible 4^R-row output is explicitly subexponential, not asserted polynomial. Any specified Boolean program h on the $2d$ bit encodings of the summaries can also be evaluated against these fibers:

$$\sum_y F_0(y)(-1)^{h(s(y))} = \sum_s W(s)(-1)^{h(s)}.$$

Its description and evaluation cost are charged, without counting its intermediate states as independent inputs. The theorem is neither globally optimal compression, unrestricted synthesis, nor sample-based statistical learning.

7 Beyond cubics, and substantive boundaries

For every fixed degree $k \geq 3$, the top k-linear additive derivative admits the same centroid construction when radical-free: there are n^2 unknowns and $O_k(n^k)$ equations. After full ANF substitution, explicitly require every monomial of degree $3, \ldots, k$ to lie within one primitive block. Top-degree structure alone does not imply this; crossed lower-degree nonlinear terms must reject this route. If the test passes, the foreground contains $O_k(n\beta^{k-1})$ monomials of fixed locality k. All matrix, Fourier, and fiber arguments remain valid, giving the same regime (5), with polynomial cost in the explicit raw ANF.

7.1 Prior machinery and an expanded guarantee

The complete interface is the contribution examined here: higher derivatives acquire blocks, rank distance handles unknown local bases, clustering and the budget DP acquire types, rounding preserves the exact residual, and the phase theorem handles that residual without an easy interaction graph. Centroid methods [2, 3], local search [4], co-clustering [5] and sum-rank geometry [6] are prior ingredients. Joint clustering and gauge synchronization also has substantial prior work; inspected recovery theorems [7] use stochastic models rather than this arbitrary symmetric finite-field kernel with adversarial error. A focused search did not locate the complete compiler, which does not establish historical first priority.

The earlier generic symmetric-rank compiler guarantees subexponential time for rank $r = o(n)$ and M scalar errors when $rM = o(n^2)$. The present promise is more structured and removes that product condition. For example, three-dimensional blocks with $c \asymp n/(3 \log n)$, a full-rank type kernel, and errors $S \sim G(n, \log n/n)$ have $R \asymp n/\log n$ and $E = \Theta(n \log n)$. They meet the present regime, while $RM = \Theta(n^2)$ is outside the older sufficient bound. This is not a proof excluding every alternative old certificate.

Nor is this a small-width promise. Lee–Lee–Oum [9, Theorem 1.1(ii)] gives this error graph rank-width $n/3 - o(n)$ with high probability. Adding a rank-R symmetric completion changes every cut rank by at most R, so the background still has rank-width $n/3 - o(n)$. Rank-width simulation [10] and active Clifford-frame simulation [11] are relevant competitors. The comparison is between guarantee parameters, not every rewrite or hybrid method.

7.2 Adaptive affine Gaussian leaves still require exponentially many pieces

For $n = 3t$, let the cubic part be $\sum_{j=1}^{t} x_{3j-2}x_{3j-1}x_{3j}$. Its third tensor is a direct sum of t volume tensors. Each has slice rank three, and slice rank is additive on direct sums [8], so its slice rank is n. If $T|_{W^3} = 0$ for a codimension-c subspace, a basis adapted to W gives slice rank at most $3c$: every nonzero term uses a complement coordinate in some slot. Thus $\dim W \le 2n/3$. Every affine leaf where the polynomial becomes quadratic has such a direction space, so any affine quadratic-leaf cover of all inputs needs at least $2^{n/3}$ leaves, including adaptive ones. Quadratic perturbations and affine disguises preserve this barrier. It does not apply to signed decompositions or all algorithms. For the phase state $2^{-n/2}\sum_x (-1)^{p(x)}|x\rangle$, the same radical-free tensor implies full Pauli stabilizer nullity: a stabilizing shift would have affine derivative and hence lie in the radical.

7.3 Small global rank does not imply few response types

The new compiler must remain alongside the previous one. Index three-coordinate blocks by all three-dimensional subspaces $W \le \mathbb{F}_2^r$, choose row bases U_W, stack them into U and put $D = UU^\top$. Then $\operatorname{rank} D = r = O(\log n)$, with $n = 3s$ and $s = \begin{bmatrix} r \\ 3 \end{bmatrix}_2$. Distinct block row orbits have sum-rank distance at least

$$\delta = \begin{bmatrix} r \\ 3 \end{bmatrix}_2 - \begin{bmatrix} r-1 \\ 3 \end{bmatrix}_2.$$

Any alignment leaves a nonzero row of $GU_A - U_B$. Only $\begin{bmatrix} r-1 \\ 3 \end{bmatrix}_2$ column subspaces lie in its kernel, and every other column block contributes a rank unit. Pairwise triangle inequalities within each approximation class therefore force total directed rank error at least $(s-c)\delta/2$ for an approximation with at most c orbits. Arbitrary diagonal blocks can absorb at most n directed rank units. At least

$$M' \ge (s-c)\delta/4 - n/2$$

off-block scalar errors remain. This is $\Omega(n^2)$ for $c = o(s)$, despite logarithmic original rank and zero original error. A volume cubic per three-coordinate block retains precisely the relevant primitive nonlinear blocks. This is a representation boundary, not counting hardness.

8 A sharp failure of finite-field Gaussian reconstruction

The general arbitrary dense quadratic background with near-linear local nonlinear perturbations remains open in this work. Replacing rational interpolation by finite-field additive phases does preserve Gaussian algebra, but loses a required correlation:

Theorem 8.1 (Joint-jet threshold)**.** *Let $F = \mathbb{F}_{2^e}$, $p_c(T) = \sum_{j=0}^{D} c_j T^j$, $D \ge 1$, and $\psi(a) = (-1)^{\mathrm{Tr}_{F/\mathbb{F}_2}(a)}$. An observation may be any complex function of Hasse jets at distinct nonzero nodes, with total jet length at most D. No linear combination of such observations, however many node sets occur, equals $\psi(p_c(0))$ for all c. Total joint length $D+1$ is sufficient and is the exact threshold.*

Proof. For one observation with jet lengths J_t, the polynomial $\prod_t (T-t)^{J_t}$ lies in its jet kernel, has degree at most D, and is nonzero at zero. A scalar multiple has trace one there. Translation by its coefficient vector preserves the observation while reversing the target sign, so its correlation with the target over all coefficients is zero. Any linear combination also has zero correlation, whereas the target's correlation with itself is $|F|^{D+1}$. Conversely Hermite interpolation with total length $D+1$ determines the polynomial. □

The populations $\mathrm{Tr}(c_0) = 0$ and 1 thus have identical full histograms for every allowed joint observation but opposite target sums. This is an observation-model result, not a lower bound on algorithms inspecting the raw description in other ways.

Other preserved failures narrow the same frontier. Groupwise Fourier dualization sends an invertible quadratic polar matrix B to B^{-1} and preserves every cross-cut rank. Gaussian moments

do not justify a sublinear perturbation cutoff: independent triples give $H(V) = (4+2V)^{n/3}$, whose every nonempty omitted tail changes the exact-count residue. For an a by a polynomial matrix $C(T)$ of entry degree at most R, nonsingular at zero, determinant multiplicities give

$$\sum_t (a - \operatorname{rank} C(t)) \le aR$$

at distinct nodes. Thus lowering a fixed full-rank cut at every node loses the small degree-to-node ratio needed by the source construction. None is a generic algorithmic lower bound.

9 Execution and limitations

The finite programs execute acquisition and exact identities; their local-phase leaves are exhaustive, not the asymptotic replacement engine. No practical speed comparison is claimed. Seven root raw-cubic cases through $n = 24$ passed 14,912 pointwise identities and all original/hidden/acquired counts; six also ran Gaussian–Walsh reconstruction. The 24-variable raw input had 979 cubic terms; acquisition returned rank 5, 75 residual edges, 29 foreground terms and signed sum -3328. The heterogeneous compiler checked its DP against finite profile enumeration and reconstructed raw cases with blocks $(5, 3, 3)$ and $(3, 3, 6)$, including 6144 pointwise checks.

Independent actual-code audits tested 27 orbit models, 3645 metric triangles, 693 two-sided gauge isometries, 27 exact medoid optima and 50,248 majority-rounding cases. Further heterogeneous tests checked 12 unknown-profile models, all four feasible budgets on another model, 14 malformed inputs, and two independently generated raw cases with 6144 pointwise identities and exact reconstructed sums -80 and 64. Five degree-$4, 5, 6$ fixtures passed 9728 pointwise identities. The three quartic fixtures ran the full orbit and Gaussian reconstruction; degree-five and degree-six cases checked centroid/locality and exact count preservation only.

The synthesis program compiled 2091 intermediate states on nine independent inputs, certified degree at most three, and acquired rank 2 with ten residual edges. Its exact zero count was 260. Across three classifiers, 192 labelled fibers and 4672 original-input labels matched independent exhaustive counts. For the nine-input program, the best function of its three acquired mod-four summaries makes exactly 190 errors among 512 inputs. This is optimality within that feature class, not accuracy on an external task. The streaming fiber iterator stores neither a character table nor output rows; finite validation wrappers optionally collect those small tables.

The strict low-rank boundary executed 21,840 gauge checks. Its largest certificate has 4185 variables, global rank 6 and 1395 primitive blocks; at most 139 response orbits force more than 387,267 off-block errors. The joint-jet failure executed 189 deficient joint-histogram equalities and three injectivity checks at the exact threshold. All proof, implementation and data files accompany this paper. Earlier passive-network sources and representation results remain unchanged.

References

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